

Year 9 Worksheets

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Abstract

1 Straight Line Graphs

Perpendicular Lines

1. Find the equation of the line passing through $(1, 4)$ perpendicular to the line $y = 3x - 1$. Give your answer in the form $ax + by + c = 0$ where $a, b, c \in \mathbb{Z}$.
2. Find the equation of the line perpendicular to $2x + 3y + 5 = 0$ passing through $(-1, 3)$. Give your answer in the form $ax + by + c = 0$ where $a, b, c \in \mathbb{Z}$.

Extension

1. Line A passes through the points $(-3, -1)$ and $(-1, 9)$. Line B passes through the points $(-2, 1)$ and $(k, 4)$. Line A and Line B are perpendicular. Find the value of k .

Answers (Extension only!)

Let m_A, m_B be the gradients of lines A and B respectively. Then

$$m_A = \frac{9 - -1}{-1 - -3} = \frac{10}{2} = 5.$$

Since B is perpendicular to A ,

$$m_B = -\frac{1}{m_A} = -\frac{1}{5}.$$

Thus B has equation

$$y = -\frac{1}{5}x + c.$$

Since B passes through $(-2, 1)$,

$$\begin{aligned} 1 &= -\frac{1}{5}(-2) + c \\ \Rightarrow c &= \frac{3}{5} \\ \Rightarrow y &= -\frac{1}{5}x + \frac{3}{5}. \end{aligned}$$

Thus when $y = 4$,

$$\begin{aligned} 4 &= -\frac{1}{5}x + \frac{3}{5} \\ \Rightarrow x &= -17 \\ \Rightarrow k &= -17. \end{aligned}$$

Can you find another method for solving this?

Gradient and Intercept

1. In each of the following identify the gradient and intercept. Your answers need not be numerical and may include variables.

(a) $y = 3x - 4$

(b) $y + 2 = -3x$

(c) $y - 7 = 3(x - 2)$

(d) $3x + 2y + 1 = 0$

(e) $y - y_1 = m(x - x_1)$

(f) $x + \pi = 3y - 2$

(g) $2x + 3y + b = ax - 4$

(h) $\frac{x}{a} + \frac{y}{b} = 1$

2. In each case, calculate the gradient between the two points.

(a) $(3, 2)$ and $(1, 7)$

(b) $(-4, -2)$ and $(6, -1)$

(c) $(\frac{1}{2}, 5)$ and $(3, \frac{2}{3})$

(d) $(-\frac{2}{5}, \frac{1}{4})$ and $(\frac{1}{2}, -\frac{7}{8})$

(e) (x_1, y_1) and (x_2, y_2)

(f) $(3a, 7a)$ and $(2a, -3a)$

(g) $(2a - 3, -2)$ and $(5a - 3, 7)$

(h) $(a, 2a^2)$ and $(5, 10a)$

2 Parabolas

Function Notation

Extension

1. Suppose we have two functions, $f(x) = x^2 + 6x + 1$ and $g(x) = \frac{1+x}{2-x}$. What is $f(g(3))$?
2. What is $g(f(x))$?

Aside: This process of combining functions in this way is known as composition of functions and we can denote the resulting function as $f \circ g$.

3. Pick any two functions f and g . Is it true that $f \circ g = g \circ f$? This property is known as *commutativity*.
4. Suppose $f(x) = \pi$ and $g(x) = x + 4$. Evaluate both $(f \circ g)(x)$ and $(g \circ f)(x)$.

Answers

1. (a) $f(g(3)) = -7$
(b) $g(f(x)) = \frac{x^2+6x+2}{1-6x-x^2}$
2. No. For example, let $f(x) = x^2, g(x) = x + 1$. Then
 $(f \circ g)(x) = x^2 + 2x + 1$ but
 $(g \circ f)(x) = x^2 + 1$.
3. $(f \circ g)(x) = \pi$
 $(g \circ f)(x) = \pi + 4$.

3 Further Factorisation

Further Factorisation

Factorise the following expressions:

1. $7(x - 2) + 3(x - 2)$

2. $5(y - 6) - 7(y - 6)$

3. $12(z - 4) - 7(4 - z)$

4. $16(2x - 4) + 5(x - 2)$

5. $3(x - 2)^3 - 2(x - 2)$

6. $(x^2 - 4x) - 4(x - 4)$

7. $y^3(7 - y) + 3y(7 - y)$

8. $5z^2(z - 3)^2 - 5(3z - z^2)^2$

Answers

1. $10(x - 2)$
2. $-2(y - 6)$ or $2(6 - y)$ for +1 style point.
3. $19(z - 4)$
4. $37(x - 2)$
5. $(x^2 + 4x + 2)(x - 2)$

For the record, the quadratic $x^2 + 4x + 2$ can be factorised but it doesn't have integer roots. It's $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$ for anyone who's interested... Go ahead and check by expanding it out!

6. $(x - 4)^2$
7. $y(7 - y)(y^2 + 3)$
8. 0